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RESEARCH INTERESTS

- **Multimodal Foundation Models for Physical AI:** Learning from vision, language, and motion for embodied understanding and action.
- **World Models and Motion Forecasting:** Modeling 3D environments, human motion, and multi-agent interactions to predict future states.
- **Prediction and Planning for Autonomous Systems:** Integrating perception, prediction, and planning for autonomous driving and robotics.
- **Efficient and Adaptive AI:** Model compression, model merging, and test-time adaptation for reliable deployment in dynamic environments.

WORK EXPERIENCE

- **DGIST**, Daegu, Korea 2025.03 – Present
Assistant Professor
- **Qualcomm**, San Diego, US (Remote) 2024.04 – 2024.09
Deep Learning R&D Intern
- **Naver Labs**, Seongnam, Korea 2021.06 – 2021.08
Research Intern

EDUCATION

- **KAIST**, Daejeon, Korea 2020.03 – 2025.02
Ph.D. in Mechanical Engineering
Advisor: Kuk-Jin Yoon
- **KAIST**, Daejeon, Korea 2018.03 – 2020.02
M.S. in Mechanical Engineering
- **KAIST**, Daejeon, Korea 2013.03 – 2018.02
B.S. in Mechanical Engineering
Double Major in Business and Technology Management

PUBLICATIONS

* equal contribution, † corresponding author

- [c17] [Learning Scene-Grounded Interaction Priors for Scene-Aware Human Motion Prediction](#)
Donghyeon Soon*, Jiil Yoon*, Hyeondong Kim, **Daehee Park**[†]
Neural Information Processing Systems (NeurIPS), 2026.
- [c16] [SRA: Spatial Reasoning Adapter via Evolving Social Interaction Graphs for Trajectory Prediction](#)
Jaewoo Jeong*, Seonkyu Song*, Hyeonwoo Park, Jegyeong Cho, Yujin Bae, Giwon Lee, **Daehee Park**,
Kuk-Jin Yoon[†]
Neural Information Processing Systems (NeurIPS), 2026.
- [c15] [Unified Prediction and Planning via Conflict-Aware Disjoint Parameter Training](#)
Taewon Seo*, Seonae Jeon*, Giwon Lee*, Kuk-Jin Yoon[†], and **Daehee Park**[†]
European Conference on Computer Vision (ECCV), 2026.
- [j2] [CascadeOcc: Rethinking 3D Occupancy World Models with Cascaded VQ Representations](#)
Kyumin Hwang*, Wonhyeok Choi*, Jaeyeul Kim, Jihun Park, **Daehee Park**[†], Sunghoon Im[†]
IEEE Signal Processing Letters, 2026.

- [c14] [EggHand: A Multimodal Foundation Model for Egocentric Hand Pose Forecasting](#)
Jaeyoung Choi*, Hyeondong Kim*, Yujin Kim, **Daehee Park**[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026.
- [c13] [BitTP: The Lightweight Trajectory Prediction Model with BitLLM for Edge-Devices](#)
Mincheol Kang*, HyunJin Lim*, Bomin Kang*, **Daehee Park**[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026.
- [c12] [Generative Active Learning for Long-tail Trajectory Prediction via Controllable Diffusion Model](#)
Daehee Park, Monu Surana, Pranav Desai, Ashish Mehta, Reuben M. V. John, Kuk-Jin Yoon[†]
IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
- [c11] [Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning](#)
Giwon Lee*, Wooseong Jeong*, **Daehee Park**, Jaewoo Jeong, Kuk-Jin Yoon[†]
IEEE/CVF International Conference on Computer Vision (ICCV), 2025. (Highlight)
- [c10] [Non-differentiable Reward Optimization for Diffusion-based Autonomous Motion Planning](#)
Giwon Lee*, **Daehee Park***, Jaewoo Jeong*, Kuk-Jin Yoon[†]
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
- [p1] [Self-Supervised 3D Occupancy Prediction with Temporal Consistency](#)
Jae-Seok Jeong, Sung-Hoon Yoon, **Daehee Park**, Kuk-Jin Yoon[†].
arXiv preprint, 2025.
- [c9] [Multi-modal Knowledge Distillation-based Human Trajectory Forecasting](#)
Jaewoo Jeong, SeoHee Lee, **Daehee Park**, Giwon Lee, Kuk-Jin Yoon[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
- [c8] [Diffusion-Guided Weakly Supervised Semantic Segmentation](#)
Sung-Hoon Yoon, Hoyong Kwon*, Jaeseok Jeong*, **Daehee Park**, Kuk-Jin Yoon[†]
European Conference on Computer Vision (ECCV), 2024.
- [c7] [T4P: Test-Time Training of Trajectory Prediction via Masked Autoencoder and Actor-specific Token Memory](#)
Daehee Park, Jae-Seok Jeong, Sung-Hoon Yoon, Jaewoo Jeong, Kuk-Jin Yoon[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- [c6] [Multi-agent Long-term 3D Human Pose Forecasting via Interaction-aware Trajectory Conditioning](#)
Jaewoo Jeong*, **Daehee Park***, Kuk-Jin Yoon[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. (Highlight)
- [c5] [Improving Transferability for Cross-domain Trajectory Prediction via Neural Stochastic Differential Equation](#)
Daehee Park, Jaewoo Jeong, Kuk-Jin Yoon[†]
AAAI Conference on Artificial Intelligence (AAAI), 2024.
- [c4] [Leveraging Future Relationship Reasoning for Vehicle Trajectory Prediction](#)
Daehee Park, Hobin Ryu, Yunseo Yang, Jegyeong Cho, Jiwon Kim, Kuk-Jin Yoon[†]
International Conference on Learning Representations (ICLR), 2023.
- [c3] [BIPS: Bi-modal Indoor Panorama Synthesis via Residual Depth-Aided Adversarial Learning](#)
Changgyoon Oh*, Wonjune Cho*, Yujeong Chae*, **Daehee Park***, Lin Wang, Kuk-Jin Yoon[†]
European Conference on Computer Vision (ECCV), 2022.
- [c2] [Unlocking the Potential of Ordinary Classifier: Class-specific Adversarial Erasing Framework for Weakly Supervised Semantic Segmentation](#)
Hyeokjun Kweon*, Sung-Hoon Yoon*, Hyeonseong Kim, **Daehee Park**, Kuk-Jin Yoon[†]
IEEE/CVF International Conference on Computer Vision (ICCV), 2021.
- [j1, c1] [Identifying Reflected Images from Object Detector in Indoor Environment Utilizing Depth Information](#)
Daehee Park, Yong-Hwa Park[†]
IEEE Robotics and Automation Letters (RA-L) and IEEE International Conference on Robotics and Automation (ICRA), 2021.

PATENTS

- 태스크 간 충돌을 완화하는 파라미터 분리 학습 기반 모델 병합 방법 및 장치
Korean Patent, Provisional application, 10-2026-0133460, 2026.07.21.
- 궤적 예측에서 거대언어모델 경량화를 위한 비트 양자화 기법
Korean Patent, 10-2026-0093495.
- 1인칭 시점 손 동작의 인지, 예측 및 제어를 위한 멀티모달 파운데이션 모델 프레임워크 및 최적화 방법
Korean Patent, 10-2025-0209315.
- Electronic device and method of identifying false image of object attributable to reflection in indoor environment thereof
United States Patent, Registered, US11282178B2.
- Dispositif électronique et procédé d'identification de fausse image ou de faux objet imputable à la réflexion dans un environnement intérieur correspondant
European Patent, Registered, EP3772701.
- 전자 장치 및 그의 실내 환경에서 반사에 의한 객체 허상을 식별하기 위한 방법
Korean Patent, Registered, KR10-2287478-0000.

POSITIONS OF RESPONSIBILITY

- Planning Committee Member, 2027 Physical AI R&D Projects, IITP, 2026.09
- Program Committee Member, IPIU 2027
- Planning Committee Member, 2026 Culture, Sports and Tourism R&D Projects, KOCCA, 2025.09

AWARDS AND HONORS

- Winner, NVIDIA Academic Grant Program, NVIDIA, 2025.12
- Outstanding Doctoral Graduate Award, KAIST ME, 2025.12
- Winner, Qualcomm Innovation Fellowship Korea (QIFK), 2024.12
- Research Scholarship, Samsung Advanced Institute of Technology, 2022.09–2025.02
- 1st Place, Best Paper Awards, IPIU 2022
- Outstanding Achievement Award, KAIST ME, 2016.09

ACADEMIC REVIEWING & SERVICE

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- IEEE/CVF International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- AAAI Conference on Artificial Intelligence (AAAI)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Intelligent Vehicles Symposium (IV)
- Winter Conference on Applications of Computer Vision (WACV)
- ACM International Conference on Multimedia (ACM MM)

- Pattern Recognition (PR)
- IEEE Transactions on Intelligent Vehicles (T-IV)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- ACM Engineering Applications of Artificial Intelligence (EAAI)

RESEARCH GRANTS

- **(PI)** 숙련자 시연데이터 기반 조선소 작업 특화 양팔 모바일 매니플레이터 및 Physical AI 작업스킬 실증 플랫폼 개발
산업통상자원부 / 한국산업기술기획평가원, 2026.08–2030.12.
- **(PI)** 경량형 온디바이스 인식 모델 최적화 연구 개발
경상북도 / 한국전자통신연구원, 2026.05–2027.12.
- 다중 휴머노이드의 자율 협업을 위한 비전-언어 모델 기반 적응형 장기 · 복합작업 계획 기술
과학기술정보통신부 / 한국연구재단, 2026.04–2030.12.
- 피지컬 AI 기반 고정밀 손기술 창작 및 서비스 기술 개발
한국콘텐츠진흥원, 2026.04–2028.12.
- 아카이브 영상 콘텐츠 활용을 위한 AI Outpainting 및 콘텐츠 변환 기술 개발
한국콘텐츠진흥원, 2026.04–2028.12.
- **(PI)** 자기 진화형 Physical LLM: 불확실한 환경에서 행동 파운데이션 모델의 강건하고 효율적인 적응 기술 연구
과학기술정보통신부 / 한국연구재단, 2026.03–2030.02.
- **(PI)** 행동 파운데이션 모델의 동적 환경으로 확장을 통한 안전하고 범용적인 인간-로봇 협동 연구
과학기술정보통신부 / 한국연구재단, 2025.09–2026.08.
- 로봇및기계전자공학연구소
교육부 / 한국연구재단, 2025.09–2034.08.
- **(PI)** Physical AI 분야의 거대 생성모델 기술 선도를 위한 LG AI STAR 인재양성 사업
과학기술정보통신부 / 정보통신기획평가원, 2025.07–2028.12.
- (InnoCORE) 초거대언어모델 혁신 연구단(DGIST)
과학기술정보통신부 / 한국과학기술원, 2025.07–2029.12.
- (InnoCORE) 바이오 체화형 피지컬 AI 연구단(DGIST)
과학기술정보통신부 / DGIST, 2025.07–2029.12.
- AI스타펠로우십지원(대구경북과학기술원)
과학기술정보통신부 / 정보통신기획평가원(IITP), 2025.04–2028.12.
- 생성형 AI 기반 16bit HDR 솔루션 개발
삼성전자-KAIST 산학과제.